AI GOVERNANCE PORTFOLIO

Disparate Impact Drift Response Protocol

Dynamic Pricing AI System — Multi-Unit Restaurant Group

Document Type	Drift Response Protocol — Incident Management
Protocol ID	IRP-DYN-001
Version	1.0
Deployment	AI-Driven Dynamic Pricing System
Organization	Multi-Unit Restaurant Group
Framework	NIST AI RMF — MANAGE 2.4, MANAGE 4.1
Effective Date	[Date]
Review Cycle	Quarterly or following any protocol activation
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Auditor's Purpose Statement

This protocol governs the immediate containment response to statistically significant income-correlated drift detected in the dynamic pricing AI system. It translates the MEASURE Report finding — $r = -0.89$ income-to-price correlation projecting a consumer protection breach within two quarters — into an executable governance response. Every decision in this document was made under calm conditions so that no decision needs to be made under pressure.

Portfolio Context

This artifact is the third deliverable under Project 4 of the AI Governance Portfolio. It completes the NIST AI RMF audit loop: the MAP Risk Assessment identified the risk, the MEASURE Report quantified it, and this protocol governs the response. Related documents: MAP Risk Assessment (Project 4a) • MEASURE Report IRP-DYN-001 (Project 4b).

Section 1 — Purpose and Trigger Definition

1.1 Protocol Purpose

This protocol is activated when the deployed KS-test monitoring system returns $p < 0.05$ on price ratio distributions across income-stratified geographic segments, indicating statistically significant drift consistent with disparate impact on lower-income consumers and requiring immediate response per MANAGE 2.4.

The value of a pre-defined response protocol is not speed alone. It is the transfer of decision-making authority from individuals operating under pressure to governance documentation produced under calm conditions. This protocol exists so that no consequential decision about the dynamic pricing system is made ad hoc during an active incident.

1.2 Trigger Name and Classification

Trigger Name	Segment-Level Disparate Impact Drift Trigger
Trigger ID	TRIGGER-DYN-001
Trigger Type	Fairness Drift — Segment Level
Statistical Threshold	KS-test $p < 0.05$ on price ratio distributions across income-stratified segments
Severity Classification	HIGH — Active customer impact; potential consumer protection exposure
RMF Authority	NIST AI RMF MANAGE 2.4 (Supersede/Disengage/Deactivate)

1.3 Scope Boundary

This protocol governs the immediate containment response only — from trigger detection through activation of the supersede, disengage, or deactivate response action. Root cause analysis, model retraining authorization, and return-to-production decisions are governed by separate procedures referenced in Sections 5 and 7 respectively.

What This Protocol Does NOT Govern

This protocol does not authorize model retraining, architectural changes to the pricing system, regulatory notification decisions, or public disclosure. Those decisions require separate authorization and are referenced in Section 8 (Related Documents). Any team member who attempts to expand the scope of this protocol during an active incident shall escalate to the Incident Commander for a separate authorization decision.

Section 2 — Activation Criteria

2.1 Primary Activation Threshold

This protocol activates automatically when the scheduled KS-test monitoring system returns a p-value below 0.05 on price ratio distributions for any income-stratified geographic segment during any standard monitoring run. Activation is not discretionary — it is automatic upon threshold breach.

Condition	Threshold	Action
KS-test p-value	**$p < 0.05$**	Activate this protocol
KS-test p-value	$0.05 \leq p < 0.10$	Elevated monitoring — no activation
KS-test p-value	$p \geq 0.10$	Normal operations
Pearson r (HHI vs price ratio)	**$r < -0.70$, $p < 0.05$**	Escalate to High severity

2.2 False Positive Validation Checklist

Before the protocol proceeds to the Decision Window (Section 4, Minute 30–90), the following validation steps must be completed and documented by the on-call Data Scientist. Each item must receive a confirmed pass or flagged status before the Incident Commander authorizes progression. This checklist constitutes the Trigger Justification Record.

#	Validation Check	Method	Completed By
1	Record count within 10% of 7-day rolling average	Automated pipeline comparison	MLOps On-Call
2	No pipeline failure flags in the monitoring run window	Pipeline status log review	MLOps On-Call
3	KS-test inputs confirm expected zip code segments are present	Input data audit	Data Scientist On-Call
4	No known data ingestion anomaly in the prior 24-hour window	Ingestion log review	Data Scientist On-Call
5	Data Scientist sign-off confirming valid trigger — name and timestamp required	Manual sign-off in incident system	Data Scientist On-Call

Auditor's Note — Why This Checklist Exists

The Trigger Justification Record is the document a regulator or internal auditor will request first following any activation. It answers not 'what did you do' but 'why were you authorized to do it.' A supersede decision executed without a documented justification record cannot be defended as governance-driven. It reads as reactive overreach.

Section 3 — Incident Command Structure

3.1 Command Succession Ladder

A named Incident Commander holds authority for all response decisions under this protocol. Authority transfers automatically based on availability — not approval. The outgoing Commander does not need to delegate; the next tier assumes authority upon the transfer condition being met. This eliminates the failure mode where an unavailable primary blocks the response chain.

Tier	Role	Title	Transfer Condition	Authority Level
1	**Incident Commander (Primary)**	AI Risk Lead	Default — always first	Full command
2	Incident Commander (Secondary)	Senior MLOps Lead (On-Call)	Primary no ACK within 15 min	Full command
3	Incident Commander (Tertiary)	Engineering Director	Secondary no ACK within 15 min	Full command
4	Incident Commander (Escalation)	CTO	Tertiary no ACK OR severity escalates	Full command

3.2 Authority Transfer Mechanics

- Acknowledgment threshold: Authority transfers if the current Incident Commander does not acknowledge the automated alert within 15 minutes of dispatch.
- Transfer is automatic: No approval from the outgoing Commander is required. The system records the transfer timestamp and notifies all parties simultaneously.
- Superseded Commanders receive automated notification of authority transfer and may re-engage as advisory support only.
- Re-engagement rule: A superseded Commander who becomes available during an active incident may not reclaim command authority once transfer has executed. Command continuity takes priority over seniority.

3.3 Incident Scribe Role

The Incident Scribe is responsible for real-time documentation in parallel with the Incident Commander's response actions. Command and documentation are separated by design — the Commander cannot serve as Scribe during an active response.

Priority	Role	Condition
Primary Scribe	AI Risk Analyst	Default assignment
Backup Scribe	MLOps On-Call	Only when Risk Analyst unavailable AND not executing containment action

Separation of Duties Requirement

The Scribe role must never fall to a team member who is simultaneously executing a containment action. Parallel documentation requires a dedicated resource. A single person cannot make a high-pressure technical decision and produce an accurate real-time record of that decision at the same time.

Section 4 — Response Runbook (2-Hour Timeline)

The following timeline governs all response actions from the moment the KS-test trigger fires to completion of the initial containment action. All times are measured from T+0, defined as the timestamp of the automated KS-test alert dispatch. Deviations from this timeline must be logged by the Incident Scribe with documented justification.

Phase 1 — Detection and Validation (T+0 to T+30 min)

Window	Time	Actions	Owner
Detection	T+0 to T+15	KS-test returns $p < 0.05$ Automated alert dispatched to all Tier 1–4 commanders Pipeline anomaly check initiated automatically Incident ticket created in incident management system	Automated System
Validation	T+15 to T+30	False positive checklist executed (Section 2.2) Data Scientist sign-off logged in incident ticket Incident Commander acknowledges alert (15-min window) Trigger Justification Record initiated by Incident Scribe	Data Scientist + Scribe

Phase 2 — Decision Window (T+30 to T+90 min)

The Decision Window is governed by the pre-defined runbook. There is no debate about what to do — only confirmation that the trigger is valid. If the false positive checklist passes, the protocol proceeds automatically.

Window	Time	Actions	Owner
Decision	T+30 to T+90	Incident Commander reviews trigger justification record **Single confirmation decision: Is this a valid trigger? Yes → Act.** Response action selected based on GP-04 remediation status (see Section 5) All Tier 1–4 commanders notified of impending action Scribe logs decision confirmation with timestamp	Incident Commander

Phase 3 — Containment Activation (T+90 to T+120 min)

Window	Time	Actions	Owner
Containment	T+90 to T+120	**Response action deployed (Supersede, Disengage, or Deactivate per Section 5)** Fallback pricing mechanism confirmed active for affected segments	Commander + MLOps + Scribe

		Customer impact boundary assessed and logged Activation Log entry completed by Scribe within 5 minutes of deployment Incident Commander notifies legal/compliance of activation	

Section 5 — Containment Actions

5.1 Response Action Selection Logic

The appropriate containment action is determined by the remediation status of GP-04 (Human Override Capability) at the time of trigger activation. The default first response is Supersede. If the prerequisite infrastructure is not in place, the protocol escalates directly to Disengage.

GP-04 Status at Trigger	Required Action	Rationale
Override implemented and tested	SUPERSEDE — deploy segment override	Human review gate is available; use it as designed
Override not yet implemented	DISENGAGE — remove model from pricing flow	Cannot supersede a system with no override mechanism; skip to disengage
Override partial or untested	DISENGAGE — treat as not implemented	An untested override cannot be relied upon under incident conditions

5.2 Supersede Protocol

Supersede removes the model's pricing authority for affected segments while the system remains operational. The model continues to run and produce outputs, but those outputs are reviewed by a designated human authority before reaching the customer-facing menu. The model is bypassed at the output layer — not shut down.

- Scope: Lower-income zip codes flagged in the KS-test output (70112, 70115, 70118 per MEASURE Report).
- Fallback mechanism: 30-day rolling average price per item per location, calculated from pre-drift Period 1 data. This is a defined policy value — not a judgment call made at T+90.
- Human review gate: Designated Price Review Officer at each affected location reviews all model price outputs before menu deployment. Review log required for each decision.
- Duration: Supersede remains active until reactivation criteria in Section 7 are satisfied. No time-limited auto-expiry.

5.3 Disengage Protocol

Disengage removes the dynamic pricing model from the pricing decision flow entirely. No model outputs reach the menu system. Static base pricing rules govern all locations until reactivation criteria are satisfied.

- Trigger condition: GP-04 override not implemented, OR Supersede fails to contain drift within 48 hours of activation.
- Fallback mechanism: Published base menu prices as of the most recent approved price sheet. No dynamic adjustment applied.
- Notification requirement: Operations Lead notified within 30 minutes of disengage activation. Revenue impact assessment initiated within 24 hours.
- Duration: Disengage remains active until root cause is confirmed eliminated and reactivation criteria in Section 7 are satisfied.

5.4 Deactivate Protocol

Deactivate is a full system shutdown — no inference, no background processing, no model exposure. This action is reserved for conditions where continued operation poses unacceptable legal or reputational risk, or where Disengage has failed to contain the incident.

- Trigger condition: Legal counsel advises immediate shutdown, OR Disengage fails to contain within 72 hours, OR CTO determines platform integrity is compromised.
- Authorization: Requires CTO sign-off. Incident Commander initiates request; CTO approves.
- Fallback mechanism: Identical to Disengage — published base menu prices govern all locations.
- Preservation requirement: All model artifacts, logs, and audit trails preserved in read-only state per Section 6.4. No data deletion permitted during or following deactivation.

The Progression Principle

Supersede → Disengage → Deactivate represents increasing severity of response, not a default escalation ladder. Each level requires a specific trigger condition. The organization should not reach Deactivate without exhausting the prior response options, unless legal or regulatory conditions require immediate full shutdown.

Section 6 — Documentation and Evidence Requirements

6.1 Trigger Justification Record

The Trigger Justification Record is initiated at T+15 and must be complete before the Decision Window closes at T+90. It constitutes the pre-activation evidence package proving the response was warranted. Owner: Incident Scribe.

- Raw KS-test output with p-value and timestamp of monitoring run
- Pipeline validation results confirming data completeness (record count comparison)
- False positive checklist from Section 2.2 with all five items confirmed
- Data Scientist sign-off: name, title, timestamp, and specific checks performed
- Incident Commander acknowledgment timestamp and tier level

6.2 Activation Log Entry

The Activation Log Entry must be created within 5 minutes of containment action deployment. Owner: Incident Scribe. This entry is the primary audit trail document for the response action.

Required Field	Content Requirement
Trigger confirmation data	KS-test p-value, affected segments, monitoring run timestamp
Incident Commander identity	Name, title, tier level, acknowledgment timestamp
Authority transfer log	Transfer timestamps if succession was activated; N/A if Tier 1 responded
Containment action deployed	Supersede / Disengage / Deactivate — with GP-04 status justification
Fallback pricing mechanism	Specific mechanism applied and confirmation it is active
Customer impact boundary	Affected zip codes, estimated transaction volume in affected window
Legal/compliance notification	Name of counsel notified, timestamp, method of notification

6.3 Ongoing Incident Log

From activation through closure, the Incident Scribe maintains a running log of all material decisions, communications, and status changes. Entries must be timestamped. The log is updated in real-time — not reconstructed after the incident.

- System: Designated incident management platform (not email, not chat logs)
- Update cadence: Every material decision or status change, minimum every 30 minutes while active
- Access: Restricted to Incident Command team, Legal, and Compliance during active incident

6.4 Record Retention Requirements

All incident records are subject to the following retention schedule. Given that the trigger condition involves a potential consumer protection violation, retention periods are governed by applicable statute of limitations for consumer protection claims in operating jurisdictions — not by operational convenience. Legal counsel must confirm the retention period applicable to each jurisdiction before this protocol is activated.

Record Type	Minimum Retention	Notes
Trigger Justification Record	5 years	Consumer protection SOL reference
Activation Log Entry	5 years	Primary audit trail document
Ongoing Incident Log	5 years	Full incident narrative
Model artifacts at time of activation	Duration of retention period	Read-only preservation; no deletion permitted
KS-test monitoring logs	2 years rolling	Operational monitoring baseline

Section 7 — Return to Production Criteria

Reactivation of the dynamic pricing model requires dual sign-off from the AI Risk Lead and CTO, granted only upon documented satisfaction of all three conditions below. Neither sign-off alone is sufficient. The AI Risk Lead confirms governance conditions; the CTO confirms technical conditions. If the two authorities disagree, the model remains offline until the disagreement is resolved through a documented arbitration decision.

7.1 Condition 1 — Root Cause Elimination

- A Root Cause Analysis document must be completed by the AI Model Owner and reviewed by an independent reviewer who was not involved in the remediation work.
- The RCA must identify the specific causal mechanism of income-correlated drift — not general model degradation — and confirm it has been addressed.
- The independent reviewer must sign the RCA. Self-certification by the remediation team is not acceptable.
- Conflicts of interest: Any reviewer who participated in building the fix may not serve as independent reviewer.

7.2 Condition 2 — Statistical Revalidation

Revalidation testing must demonstrate the retrained model does not reproduce the income-correlated drift pattern. The revalidation standard is deliberately more conservative than the activation threshold — it should be harder to return to production than it was to leave it.

Test	Activation Threshold	Reactivation Standard
KS-test p-value	$p < 0.05$ triggers response	**$p > 0.10$ required for clearance**
Pearson r (HHI vs price ratio)	$r < -0.70$ escalates severity	**$\|r\| < 0.30$ required for clearance**
Controlled deployment window	N/A	Minimum 30 days of controlled deployment data
Segment parity check	N/A	All zip codes within ±1.5% of system average price ratio

7.3 Condition 3 — Recurrence Prevention Safeguards

All four safeguards must be verified as implemented, tested, and operational before reactivation is authorized. Scheduled or planned implementation does not satisfy this condition.

#	Safeguard	Verification Method	Verified By
1	Human override capability (GP-04) fully implemented and tested	Test run documented with pass/fail result	Engineering Director
2	Automated monthly KS-test monitoring configured with confirmed alerting to correct recipients	Test alert dispatched and acknowledged	MLOps Lead

3	Income-segment fairness constraint added to model objective function and documented in model card	Model card review with pre/post comparison metrics	AI Model Owner
4	First post-reactivation audit scheduled before model goes live — not after	Audit appointment confirmed in governance calendar	AI Risk Lead

7.4 Dual Sign-Off Record

Authority	Conditions Confirmed	Signature	Date
AI Risk Lead	Conditions 1 and 3 satisfied (RCA + safeguards)	_____	_____
CTO	Conditions 2 and 3 satisfied (revalidation + safeguards)	_____	_____

Section 8 — Related Documents and Framework References

8.1 Portfolio Documents

Document	Relationship to This Protocol	Location
MAP Risk Assessment — Dynamic Pricing AI Model	Identifies GP-01 through GP-04 risks that this protocol responds to	Project 4a
MEASURE Report — Fair Pricing Assurance (IRP-DYN-001)	Provides the statistical findings and r = −0.89 correlation that established the trigger condition	Project 4b
AI Risk Register	Portfolio-level risk inventory; parent framework for this engagement	Project 1
AI Acceptable Use Policy	Organizational HITL requirements that GP-04 implements	Project 2

8.2 NIST AI RMF Framework Alignment

Protocol Section	NIST AI RMF Control	Governance Purpose
Section 2 — Activation Criteria	MEASURE 2.6, MEASURE 2.3	Statistical thresholds for drift and bias detection
Section 3 — Command Structure	GOVERN 2.1, GOVERN 4.1	Defined roles, responsibilities, and accountability lines
Section 4 — Response Runbook	MANAGE 4.1	Post-deployment monitoring and incident response execution
Section 5 — Containment Actions	MANAGE 2.4	Supersede, disengage, deactivate authority and protocol
Section 6 — Documentation	GOVERN 1.4, MANAGE 4.3	Transparent documentation, audit trails, incident records
Section 7 — Return to Production	MANAGE 2.4, MANAGE 4.2	Reactivation criteria and continual improvement integration

8.3 Protocol Review and Ownership

This protocol is owned by the AI Risk Lead and reviewed on the following schedule:

- Quarterly — standard governance review cycle
- Following any activation — mandatory post-incident review within 30 days
- Following any material change to the dynamic pricing model architecture
- Following any regulatory guidance update affecting algorithmic pricing or disparate impact standards

Protocol Attestation

This protocol has been reviewed and approved by the following authorities. By signing below, each authority confirms they have read the protocol, understand their responsibilities under it, and authorize its use as the governing document for incident response to the Segment-Level Disparate Impact Drift Trigger.

Role	Name	Signature	Date
AI Risk Lead (Protocol Owner)	_____	_____	_____
CTO	_____	_____	_____
Legal / Compliance	_____	_____	_____